Competency 13: Explores new trends and future directions of ICT

Competency Level 13.1 Explores new trends and future directions in computing

Learning Outcomes:

- Describes intelligent and emotional computing.
- Explains artificial intelligences
- Appreciates man- machine coexistences

New Trends and Future Directions of ICT

Nowadays ICT is changing our lives more than ever before. The speed and convenience of many of our day today activities rely on ICT. This heavy adoption of ICT by both individuals and markets has inspired scientists and engineers in ICT to actively do research on new information and communication technologies as well as their applications in various domains. The Gartner hype cycle[32]– 2019 presented in Figure 5.1 depicts the state of adoption, maturity and social application of many emerging technologies that are related to ICT as of 2020. The hype cycle, which is a very useful tool to recognize top emerging technologies in the world, has five key phases that shows;

1. The inception or the trigger of a technology
2. Peak with inflated expectations
3. Trough of disillusionment with failed implementations and promises
4. The slope of enlightenment with success stories
5. Plateau of productivity with mainstream adoption

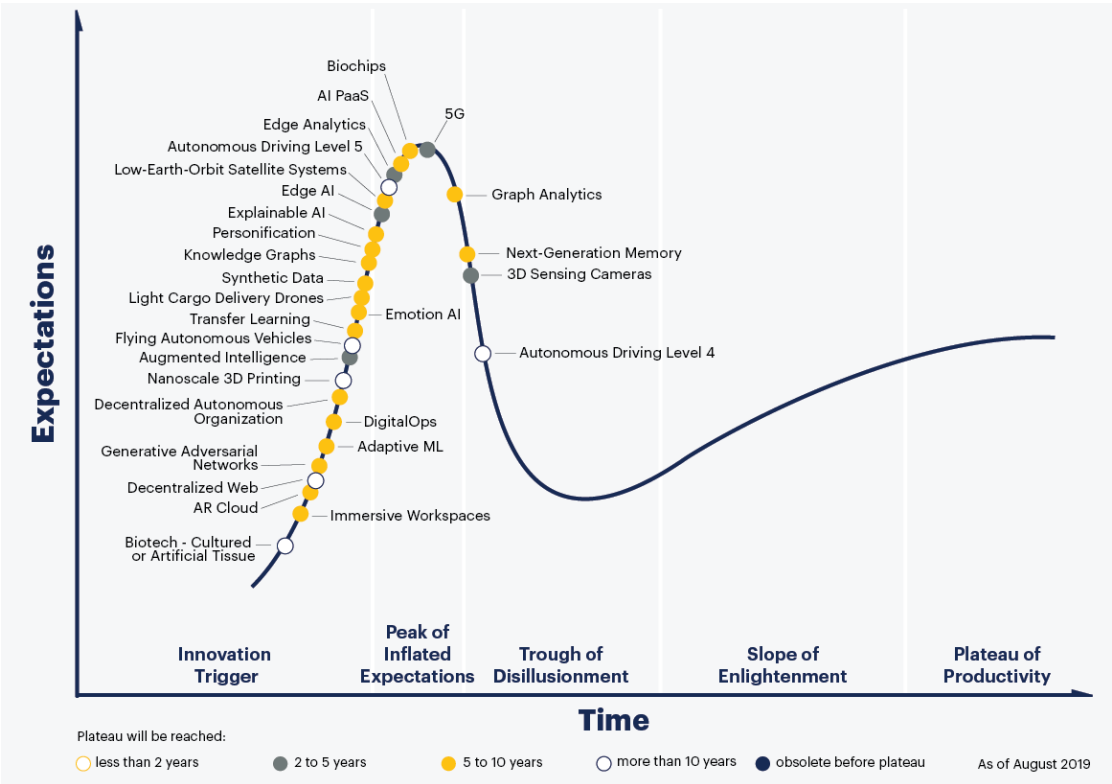

Figure 5.1: Gartner hype cycle – 2019
Source: https://towardsdatascience.com/gartner-2019-hype-cycle-for-emerging-technologies-whats-in-it-for-ai-leaders-3d54ad6ffc53

[32] Read https://en.wikipedia.org/wiki/hype_cycle

The notable emerging and trending information and communication technologies mentioned in the hype cycles of 2018 and 2019 include Artificial Intelligence, Blockchain, IoT platforms, Quantum Computing and smart robots. Moreover, they mention about many other technologies that involve artificial intelligence as the core technology such as deep neural networks, autonomous driving, knowledge graphs, flying autonomous vehicles, brain-computer interface, etc. Thus, artificial intelligence, which leads to the creation of intelligent and emotional machines that help mankind in various endeavors, could be considered as one of the key emerging information and communication technologies.

Intelligent and emotional machines

Evidence on human interest about intelligent and emotional machines could be found even in the ancient Greek mythology, which contain characters such as Talos and Caucasian Eagle sent to earth by Zeus for human protection. Even the modern fictions such as '2001: A Space Odyssey' by Sir Arthur C Clarke as well as the famous 80's tele series 'Knight Rider' had machines that exhibit not only intelligent but also some emotional characteristics. For example, KITT, the intelligent car in Knight Rider, engages in several conversations with its master Michael Knight that shows some emotions. On the other hand, humans, throughout history, have attempted to build self-operating machines as well as machines with control mechanisms designed to automatically follow a predetermined sequence of operations, or respond to predetermined instructions. Those machines are known as automatons. However, the growth of such attempts to build truly intelligent machines had been very slow until the discipline of artificial intelligence was formally begun in the mid-20th century. Figure 5.2 contains the famous Maillardet's automaton found in Franklin Institute's museum.

Figure 5.2: The famous Maillardet's automaton in Franklin Institute's museum
Source: https://www.fi.edu/history-resources/automaton

Artificial intelligence

The term artificial intelligence could be considered as coined during the 1956 summer workshop held at the Dartmouth College in Hanover, New Hampshire. There, a group of researchers interested in thinking machines did an extended brainstorming session to elicit innovative ideas on creating such machines. Since then, various definitions of artificial intelligence have been proposed by scientists. According to one such definition, AI (artificial intelligence) is the simulation of human intelligence processes by machines, especially computer systems[33]. The human intelligence processes include the ability to perceive the environment, synthesis perceived

[33]https://searchenterpriseai.techtarget.com/definition/AI-Artificial-Intelligence

information, learn and remember facts and apply knowledge to reason about events and solve problems. In general, the intelligence demonstrated by machines so far is considered only as imitating human (or natural) intelligence. This could be well explained by the two hypotheses presented by John Searle in 1980. According to Searle, artificial intelligence has two forms; strong form and the weak form.

- Artificial intelligence, strong form: an AI system can think and have a mind
- Artificial intelligence, weak form: an AI system can only pretend it thinks and has a mind

Thus, in order to create a truly intelligent and emotional machine, one must focus on the strong form of artificial intelligence. The researchers in the field of Artificial General Intelligence (AGI) are working on the creation of such machines (or computer systems) which are capable of working in multiple fields. Such a machine, for example, would learn to drive a car (one field) as well as defuse a landmine (another field). Moreover, machines with a strong form of AI demonstrate emotions since they have a mind. For example, such machines may learn to appreciate (truly) an object in terms of its beauty! There is another extension of the strong form of artificial intelligence called Artificial Super Intelligence, which is based on the idea that a system with the capabilities of an AGI, in the absence of the physical limitations of humans, would learn and improve far beyond the human level as a super system.

However, humans have yet been able to develop intelligent machines only with the weak form of artificial intelligence. That means, such machines demonstrate intelligence similar to humans (or even better than humans) when performing some narrow tasks such as playing a board game or predicting weather. However, their capabilities are limited only to that narrow task and their ability to improve their capabilities on other tasks is minimal. Therefore, such a form of artificial intelligence is also called artificial narrow intelligence. IBM Deep Blue that defeated the world champion Garry Kasparov in the game of Chess (1997), IBM Watson that defeated human players in a gameshow called Jeopardy (2011), and the algorithm called AlphaGo that defeated the human champion in the game of Go (2016) could be considered as machines with weak intelligence.

There are several techniques of artificial narrow intelligence developed by the artificial intelligence researchers to use when building intelligent systems and some notable techniques among them are mentioned in Table 5.1.

Man-machine coexistence

One of the key fears of mankind when it comes to the creation of intelligent machines is that they will one day take the place of humans. Thus, the research in artificial intelligence as well as ICT in general emphasis the need for man-machine coexistence in which humans and machines work together to create unprecedented breakthroughs in the workforce[34]. For example, the Honda Research's humanoid robot Asimo was developed to work with humans in workplaces corporatively[35]. Asimo has also demonstrated its ability to work with multiple other Asimos, which is an example for the concept of machine-machine coexistence.

[34] Read https://www.forbes.com/sites/forbestechcouncil/2017/07/06/man-machine-and-multiplicity-how-ai-and-humans-can-coexist-harmoniously/#524d497317df

[35] Watch video https://www.youtube.com/watch?v=JlRPICfnmhw

Technique	Description
Search techniques	Searching for a goal state in state space. For example, finding the winning state in a board game with a human player
Expert systems	Rule-based systems that store knowledge as If-Then rules and advice humans to solve problems with reasoning using the stored knowledge.
Natural Language Processing	Algorithms that recognize and even understand human languages
Speech recognition	Algorithms that recognize words in speech and understand short voice sentences
Computer vision and scene recognition	Algorithms that interpret and understand visual contents in scenes captured by cameras
Machine learning	A set of techniques that can learn hidden patterns in data. For example, machine learning algorithms can be used to predict whether a new customer will default a bank loan or not by exploring the past loan data of all customers of the bank
Neural networks	A notable technique in the domain of machine learning, which is based on a large network of information processing units called artificial neurons which are organized into two or more layers. A significant development in this area could be seen with the emergence of deep neural networks
Genetic algorithms	A class of optimization algorithms based on the biological process of evolution. There, similar to the survival of the fittest rule in biological species, current good (fit) solutions are combined to generate better solutions until a significantly optimal solution is obtained.
Fuzzy logic systems	A technique used to control systems smoothly through fuzzy rules based on linguistic statements. For example, if the room temperature is ‘hot’, turn the fan ‘fast’ is a fuzzy rule. Here ‘temperature’ and ‘fan speed’ are fuzzy variables that take fuzzy values like ‘hot’ and ‘fast’ rather than binary values ‘true’ or ‘false’

Table 5.1: Intelligent Techniques

Competency Level 13.2 Explores the fundamentals and applications of agent technology

Number of Periods: 04

Learning Outcomes:

- Briefly describes software agents and their characteristics
- Briefly describes multi-agent systems and their characteristics
- Identifies the applications of agent systems

Software agents

In computer systems, a software agent is a piece of software that works on behalf of a human user or another computer application. Cortana, in Windows could be considered as an example for a software agent. It is a piece of software inside the Windows operating system that acts as a virtual assistant. It provides personalized services to the human user such as searching for apps and files, recognizing voice commands, giving reminders, etc. Software agents in most cases reflect (weak) intelligence by their abilities such as recognizing voices, interpreting the text in natural language, predicting based on past data, and reasoning based on observations. Software agents could be:

- **Autonomous:** can take decisions by themselves without the interference of the human user
- **Proactive:** can initiate actions on its environment
- **Reactive:** can react to events in the environment
- **Cooperative:** can cooperate with other agents and humans
- **Able to learn:** can learn by observation, experience, and data
- **Social ability:** can interact and communicate with others complying to the social rules and norms

These features actually help to differentiate a software agent from an object in software. A good classification of software agents is depicted in Figure 5.3.

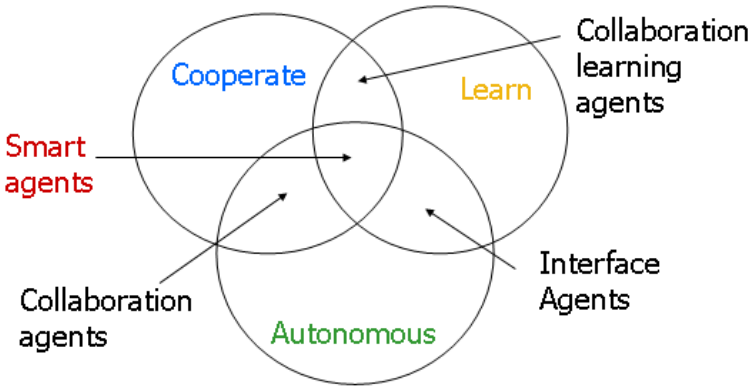

Figure 5.3: a classification of software agents based on Nwanna's primary attribute dimension
Source: https://en.wikipedia.org/wiki/Software_agent#/media/File:Nwana_Category_of_Software_Agents.gif

Multi Agent Systems (MAS)

Software agents can interact with other software agents to execute a common task. In most cases, a single software agent is not capable of solving complex problems. Thus, the applications aiming at solving complex problems require multiple software agents working as a single unit, which is called a Multi-Agent System or MAS. In other words, a multi-agent system (MAS) is a

loosely coupled network of software agents who interact to solve problems that cannot be solved by individual problem solvers[36]. In such a multi-agent system, the agents are[37];

- **autonomous:** meaning that they can make independent decisions by being self-aware
- **having only a local view:** meaning that they only know how to execute their part and cannot see beyond their scope
- **acting in a decentralized manner:** meaning that there is no agent to control the system

Figure 5.4 depicts a general functionality of a multi-agent system in which a user gets some information from multiple sources through an information broker. Each source is assigned to an information agent who submits the relevant information to the broker who filters and provides the appropriate information to the user. The user accesses the system through an interface agent.

Applications of Agent systems

Agent systems are found in a range of applications. As mentioned before Cortana in Windows as well as Siri on Apple iOS systems are good examples for software agents that are practically implemented. Such virtual assistants could be seen in many computer systems to make the tasks of the human users easier. Agents also assist humans in banking, e-learning, e-commerce, booking, and in various other similar systems. For example, information searching in electronic commerce applications involves a multi-agent system similar to the architecture depicted in Figure 5.4. There, the users submit their search criteria for products to an interface agent. The interface agent then connects with an information broker agent who filters information received through agents at various merchants' websites in response to the users' search criteria to provide the relevant information to the users. Online booking systems too adopt such architecture to filter information on behalf of the users, connect to respective merchants and do the booking on behalf of the human user.

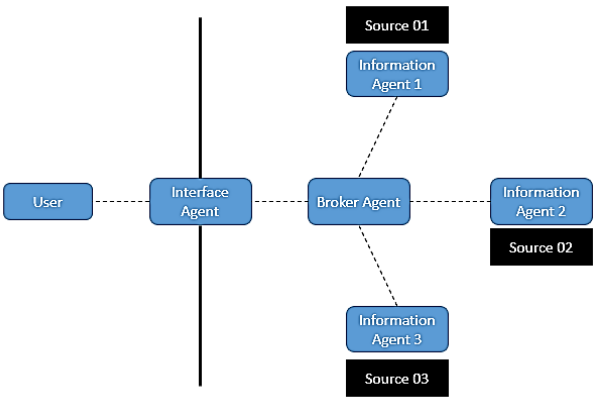

Figure 5.4: Architecture of a multi-agent system

[36]https://www.cs.cmu.edu/~softagents/multi.html
[37]https://en.wikipedia.org/wiki/Multi-agent_system

Competency Level 13.3 Analyzes the existing models of computing and proposes new models

Learning Outcomes:

- Predicts the technologies beyond von-Neumann computers

Beyond Von-Neumann computer

The von Neumann architecture is the basis for the modern computer. As depicted in Figure 5.5, it provides a simpler architecture to design computer systems. However, as the demand for computational resources rapidly grows, the von Neumann computer has now been pushed to its extreme limits of capabilities. According to Moore's law, the processing speed of microprocessors doubles once every 18 months. However, there are physical limitations for this trend to continue further such as managing the excessive heat generated in microprocessors when they are functioning with more and more transistors embedded into them to increase the processing speed. Therefore, the researchers have already started to look for alternative architectures that can meet the future demand for computational resources conveniently.

Quantum computing

Quantum computing is an alternative approach proposed to overcome the limitations of the modern-day microprocessors. In the traditional approach, information is represented using a binary system which has only two states (0 and 1). However, quantum computers which are based on quantum physics, can represent information using multiple states. Therefore, quantum computers would be exponentially faster than the traditional computers based on the von Neumann architecture[38].

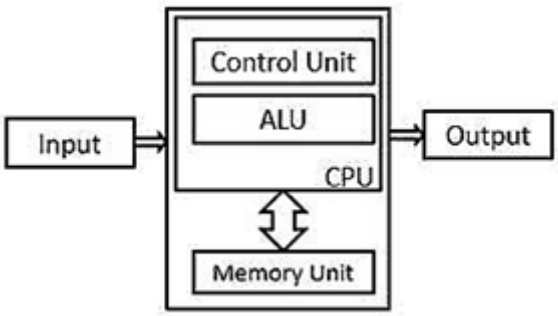

Figure 5.5: von Neumann architecture
Source: http://www.polytechnichub.com/wp-content/uploads/2017/04/Von-Neumann-architecture.jpg

Nature-inspired computing

Researchers are inspired by nature when developing new computing techniques, especially intelligent computing techniques. They observe natural phenomena and try to understand how such phenomena could be helpful to address complex natural problems. Thereby the researchers become able to develop new techniques that mimic such natural processes to solve complex problems that mankind faces nowadays. Such techniques are generally called nature-inspired computing techniques or natural computing techniques.

In nature-inspired computing, for example, the researchers observe how biological species, like ant colonies, beehives, and flocks of birds, react to stimuli, process information and make decisions. For example, certain flocks of birds collectively display altruistic behavior by leaving

[38]https://whatis.techtarget.com/definition/quantum-computing

a healthy bird to remain with an injured bird on the ground[39]. Such observations have given rise to groundbreaking computing techniques such as neural networks, swarm intelligence, evolutionary computation and artificial immune systems. This branch of computer science is closely related to biology-inspired computing and artificial intelligence[40].
Table 5.2 contains some additional nature-inspired computing techniques other than genetic algorithms and neural networks that have been introduced before.

Technique	**Description**
Swarm intelligence	An artificial intelligence technique based on the intelligent collective behaviors observable in the entities comprising of swarms of creatures such as ant colonies, beehives and bird-flocks. For example, bees are observed to have optimally selected (i.e. collective intelligence) the location for their hive in most of the time. Swarm intelligence enables to model complex and adaptive systems such as social systems as collections of agents who interact with each other based on some simple rules. The emergent patterns resulting from those individual agents' interactions are used to understand and control the particular system. Examples of swarm intelligence include developing optimal routing strategies, simulating crowd movements, building teams, etc.
Membrane computing	A branch of natural computing that abstracts computing models from the architecture and the functionality of living cells, as well as from the organization of cells in tissues, organs (brain included) or other higher-order structures such as colonies of cells (e.g., of bacteria)[41].
Intelligent communication systems and protocols	Observing phenomena such as the patterns of spreading epidemics to develop new communication systems and protocols, which are faster and robust

Table 5.2: Nature inspired computing techniques

Apart from the above techniques, creating sensor networks by observing the sensory organs, creating artificial immune systems by observing the biological immune systems and making learning classifier systems based on human cognition are examples for nature or biology-inspired computing[42].

[39]https://www.computersciencedegreehub.com/faq/what-is-nature-inspired-computing/
[40]https://en.wikipedia.org/wiki/Bio-inspired_computing#Areas_of_research
[41]http://www.scholarpedia.org/article/Membrane_Computing#Applications
[42]https://en.wikipedia.org/wiki/Bio-inspired_computing#Areas_of_research

Review Questions

1. Discuss the intelligent features of the machine called KITT in the tele series 'Knight Rider'
2. Present your views on the limitations of intelligence of the chess playing machine called IBM Deep Blue
3. Explain the functionality of intelligent agents in online auction, e-learning and intrusion detection systems using the common characteristics of intelligent agents
4. Explain the reasons for the current high interest about machine learning and neural networks
5. Discuss how intelligent agent technology could be used when creating information systems for the following
    a. Managing natural disasters such as flood and tsunami
    b. E-commerce
    c. Banking activities
6. Explain why interface agents are required when creating agent systems
7. Discuss how agent and multi agent technologies could have been used in the robot called Asimo
8. Explain why we need an architecture that has gone beyond the Von-Neumann architecture
9. Suppose an expert system has been created to assist a professional who provides education-counselling service to students.
    a. How should the functionality of such a system be?
    b. Is it possible to consider this system as an example for an agent system? Justify your answer
10. Explain with examples why it is important to study natural phenomena in order to create intelligent machines